

Indent Serial No.

Date:

**UNIVERSAL TRIBOMETER LABORATORY
CENTRAL RESEARCH FACILITY
IIT (ISM), DHANBAD – 826004**

[Please read the information given overleaf before filling up this form and put a tick (✓) in the appropriate box]

1. I wish to get _____ (in words _____) number of samples be measured on hourly basis.
2. The nature of samples is POLYMER/RUBBER (), METTALIC (), or OTHERS (); if others please elaborate:
3. I wish to study my sample/s as per ASTM _____ standard and using the following settings:

	Reciprocating Setup	Rotary Setup	Scratch test
Load [3-200N]			
Frequency [1-20] Hz			-
Time – second			
Stroke Length (mm)	5 mm	-	

4. I need to analyze the sample by using a system-integrated profilometer. I am interested in:
WLI: (), Confocal: ().

Please allot measurement slots ofhour/s/ (as per Annexure - I)

User's Name (block letter):

Name of Supervisor

Dept./Centre

Cell No:

Signature of HOD/HOC/Guide/PI/Prof.-in-Charge

Signature of the User

Please allot time and complete the work.

Signature of the Laboratory In-Charge

The above work has been done satisfactorily on _____(Date) within _____ number/s of slot/sample and generated data has been delivered to me.

Signature of the Operator

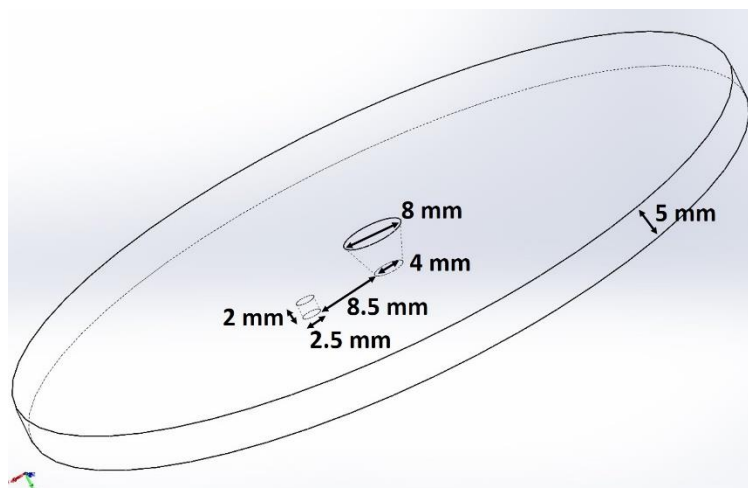
Signature of the User

INFORMATION FOR UNIVERSAL TRIBOMETER USERS

- 1) The Universal Tribometer is available for booking every week: Total of 3 sessions (Monday-Thursday-Friday). Each session is for two hours (including sample mounting, testing, and data collection time).
- 2) One full slot is of 2-hour duration.
- 3) Only one booking is allowed to a User from a research group/PI/Guide from the various department in the institute. The user from the same research group has to wait until the next round of new bookings for the next slot.
- 4) User must provide good quality of sample surface for better data quality and non-damage of the machine.

5) Dimension required for lower sample tribological test are as follows:

- a) Reciprocating: Maximum – 30 X 30 X 5 mm and Minimum - 10 X 10 X 5 mm (For Dry, wet, and Temperature controlled test). Samples should be provided in phenolic mould for better holding.
- b) Rotary: For Dry run: Diameter – 20 to 110 mm, Thickness – 5mm with tapered center hole with upper diameter of 8 mm and lower diameter of 4 mm, chamfered in 45 deg along with a blind hole of 2.5 mm diameter and 2 mm length from bottom side at a distance of 8.5 mm from the central hole. Sample image attached.



For Lubricant and Temperature controlled test: Maximum Diameter - 50 mm other dimension same as above.

- 6) Ball size available: 3, 4, 6, 10 mm and pin diameter of 3, 4, 6, 10 mm can be fitted.
- 7) **Only balls will be provided during the tests; users should bring all other samples.**
- 8) The nature samples should be declared correctly.
- 9) The deadline for submitting slot booking forms is Thursday (up to 3 PM) of the previous week.
- 10) Allotment of slots will be notified at 4 PM the next day (i.e., Friday).
- 11) Tribometer will run on Reciprocating module and rotary module on a week-to-week basis.
- 12) All forms must be forwarded through the concerned HOD, HOC, PI, Guide or Prof.-in-Charge, etc., and to be submitted in the Universal Tribology Laboratory, Central Research Facility.

- 13) In case of power shut down or unforeseen disturbances, any unutilized slot will be subsequently compensated in the next round of booking.
- 14) All users should inform composition and sample details to the operator and leave a copy of the data/results in the laboratory for internal records (log book).
- 15) Generated data will be provided to the user only through the new, unused blank CD/DVD. No request will be entertained for providing generated data by e-mail, pen drive, external hard disk, or any other means. Thus, the user may bring a new CD for the collection of their data.